Yeonsu Chang

Integrated Ph.D Candidate

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Graph Theory | Structural Graph Theory | Parameterized Complexity | Reduced Parameter

Research Statement

My research focuses on structural aspects of graphs and related width parameters, with emphasis on characterizations and algorithmic consequences in parameterized complexity.

Education

Integrated Ph.D in Mathematics Mar 2022 – Present

Hanyang University, Seoul, South Korea
Advisor: O-joung Kwon

B.S. in Mathematics Mar 2015 – Feb 2022

Hanyang University, Seoul, South Korea (incl. two years of military service)

Military Service

Republic of Korea Army Jan 2017 – Oct 2018

Completed mandatory military service (Sergeant)

Journal Papers

3. **Pre-assignment problem for unique minimum vertex cover on bounded clique-width graphs**
Shinwoo An, Yeonsu Chang, Kyungjin Cho, O-joung Kwon, Myounghwan Lee, Eunjin Oh, Hyeonjun Shin.
Theoretical Computer Science 1084 (Sep, 2026), 116173, *AAAI2025 accepted*, arXiv:2408.09591

2. **A new width parameter of graphs based on edge cuts: α-edge-crossing width**
Yeonsu Chang, O-joung Kwon, and Myounghwan Lee.
Discrete Applied Mathematics 380 (Feb, 2026), *WG23 accepted*, arXiv:2302.04624

1. **A characterization of graphs of radius-r flip-width at most 2**
Yeonsu Chang, Sejin Ko, O-joung Kwon, and Myounghwan Lee.
Discrete Mathematics 348(4) (Apr, 2025), 114366, arXiv:2306.15206

Refereed Conference Papers without Journal Version

1. **Moderately beyond clique-width: reduced component max-leaf and related parameters**
Édouard Bonnet, Yeonsu Chang, Julien Duron, Colin Geniet, and O-joung Kwon.
ESA 2026 accepted, arXiv.2604.19138

Preprints

3. **QPTAS for Max Weight Independent Set in Graphs with Strongly Edge-Sublinear Separators**
Édouard Bonnet and Yeonsu Chang.
arXiv.2610.16025

2. **Max Independent Set Remains NP-hard when Excluding a Planar Induced Minor**
Édouard Bonnet and Yeonsu Chang.
arXiv.2609.11285

1. **The optimal chromatic bound for even-hole-free graphs without induced seven-vertex paths**
Yeonsu Chang, Shenwei Huang, and Yidong Zhou.
arXiv:2602.04403

Talks

MC2 Seminar at ENS de Lyon Sep 24, 2026

Reduced Component Max-Leaf: Applications and Further Results

European Symposium on Algorithms 2026 (ESA 2026) Aug 31– Sep 2, 2026

Moderately beyond clique-width: reduced component max-leaf and related parameters

ASIACOMB 2026 Aug 24– 28, 2026

Algorithmic Applications of Reduced component max-leaf

Korean Mathematical Society 80th Anniversary Conference Jun 22– 25, 2026

The optimal chromatic bound for (even-hole, P_7)-free graphs

Korean Student Combinatorics Workshop 2026 Winter (KSCW2026W) Feb 2– 6, 2026

Structural and Algorithmic properties of Reduced Component Max-Leaf

2025 Korean Mathematical Society Annual Meeting Oct 22– 24, 2025

Structural and Algorithmic properties of Reduced Component Max-Leaf

Math Colloquium at Hanyang University Nov 4, 2024

Combinatorial game and graph structure: cops and robber game and flipper game

2023 KMS Annual Meeting Oct 26– 28, 2023

A characterization of graphs of radius-r flip-width at most 2

Teaching Experiences

Linear Algebra 1, Hanyang University Spring 2025

Math Capstone PBL (Solving Graph Problems), Hanyang University Fall 2024

Linear Algebra 1, Differential Geometry, Hanyang University Spring 2024

Head TA, Hanyang University Fall 2022 – Spring 2024

Math Capstone PBL (Applications of graph algorithms), Hanyang University Fall 2022

Linear Algebra 1, Hanyang University Spring 2022

Workshops and Conferences (In Korea)

2026 Summer School on Combinatorics and Algorithms Aug 10– 14, 2026

KAIST, Daejeon, South Korea

2026 Korean Student Combinatorics Workshop (KSCW 2026) Jul 27– 31, 2026

Gongju, South Korea

37th KIAS Combinatorics workshop May 28– 30, 2026

Jeju, South Korea

36th KIAS Combinatorics workshop Mar 13– 14, 2026
KIAS, Seoul, South Korea

2025 Korean Student Combinatorics Workshop Winter (KSCW 2025W) Feb 2– 6, 2026
Yeosu, South Korea

Workshop on Algorithm for Student 2026 Jan 26– 28, 2026
POSTECH, Pohang, South Korea

35th KIAS Combinatorics workshop Dec 22– 24, 2025
Busan, South Korea

2025 KMS Fall Meeting Oct 22– 24, 2025
Seoul, South Korea

2025 Korean Student Combinatorics Workshop (KSCW 2025) Aug 20– 24, 2025
Gyeongju, South Korea

2025 Combinatorics Workshop Aug 18– 20, 2025
IBS, Daejeon, South Korea

2025 Summer School on Combinatorics and Algorithms Jul 14– 17, 2025
POSTECH, Pohang, South Korea

34th KIAS Combinatorics workshop May 29– 31, 2025
Jeju, South Korea

2025 KMS Spring Meeting Apr 24– 26, 2025
KAIST, Daejeon, South Korea

IBS-DIMAG Winter School on Graph Minors Feb 3– 19, 2025
IBS, Deajeon, South Korea

One Day Workshop on Graph Theory Jan 15, 2025
Ajou University, Suwon, South Korea

The 4th Korean Logic Day Jan 13– 15, 2025
Changwon, South Korea

32th KIAS Combinatorics Workshop Dec 19– 21, 2024
Busan, South Korea

2024 KMS Fall Meeting Oct 24– 26, 2024
Sungkyunkwan University, Suwon, South Korea

2024 Korean Student Combinatorics Workshop (KSCW 2024) Jul 29– Aug 2, 2024
Gongju, South Korea

2024 Summer School on Combinatorics and Algorithms Jul 22– 26, 2024
KIAST, Daejeon, South Korea

Small workshop on parameterized complexity Apr 30– May 1, 2024
POSTECH, Pohang, South Korea

30th KIAS Combinatorics workshop Mar 8– 9, 2024
KIAS, Seoul, South Korea

29th KIAS Combinatorics workshop Dec 26– 28, 2023
Busan, South Korea

2023 KMS Fall Meeting Oct 26– 28, 2023
Seoul National University, Seoul, South Korea

2023 Combinatorics Workshop Aug 16– 18, 2023
Yonsei University, Seoul, South Korea

2023 KIAS Summer Workshop on Graph Theory Aug 8– 9, 2023
KIAS, Seoul, South Korea

Lecture Series on Combinatorics at Hanyang University Aug 1– 2, 2023
Hanyang University, Seoul, South Korea

28th KIAS Combinatorics workshop Jun 1– 3, 2023
Jeju, South Korea

Seminar on Combinatorial Algorithms in Hanyang University Apr 4– 5, 2023
Hanyang University, Seoul, South Korea

27th KIAS Combinatorics workshop Feb 23– 24, 2023
KIAS, Seoul, South Korea

Workshop on Graph Coloring and Related Topics Jan 28– 29, 2023
Konkuk University, Seoul, South Korea

26th KIAS Combinatorics workshop Dec 20– 22, 2022
Busan, South Korea

2022 combinatorics workshop Sep 30– Oct 1, 2022
GIST, Gwangju, South Korea

2022 School on Structural Graph Theory in Hanyang University Sep 16– 17, 2022
Hanyang University, Seoul, South Korea

Combinatorics Incheon Workshop on Jeju Jul 25– 29, 2022
Jeju National University, Jeju, South Korea

25th KIAS Combinatorics workshop Jun 15– 18, 2022
Jeju, South Korea

Workshops and Conferences (International)

GROW 2026 Sep 27– Oct 1, 2026
Będlewo, Poland

ALGO 2026 Aug 31– Sep 4, 2026
L'Aquila, Italy

ASIACOMB 2026 Aug 24– 28, 2026
Daejeon, South Korea

International Conference for the 80th Anniversary of the KMS Jun 22– 25, 2026
Seoul, South Korea

Focused Workshop on Erdős–Pósa Problems Mar 15– 20, 2026

Będlewo, Poland

SIAM Conference on Discrete Mathematics (DM24) Jul 8– 11, 2024
Spokane, Washington, U.S.

3rd Korea-China Young Scholar's Symposium on Discrete Mathematics Jun 24– 27, 2024
Jeju, South Korea

WG 2023 – Graph-Theoretic Concepts in Computer Science Jun 28– 30, 2023
University of Fribourg, Fribourg, Switzerland

Research Visits

ENS de Lyon, Lyon, France (Host: Édouard Bonnet) Sep 6– 25, 2026

Nankai University, Tianjin, China (Host: Shenwei Huang) Sep 21– Oct 1, 2025

Academic Services

- Co-organizing 2026 Korean Student Combinatorics Workshop (KSCW 2026, 2026 조합론 학생 워크샵)

Awards

Best Paper Award. "A characterization of graphs of radius-r flip-width at most 2" Feb, 2026
Graduate Student Paper Competition, Institute of Natural Sciences, Hanyang University